Analytical Cume (October 27, 2000)
Theme: Electrochemistry and Quantitative Analysis

1. ELECTROPLATING (50 POINTS)

(a) (15 points) A cobalt/tin plating bath at an electroplating company contains both Co^{2+} and Sn^{2+}. Their reduction potentials are -0.28 vs. NHE and -0.14 vs. NHE, respectively. Approximately what potential (vs. NHE) or potential region would you need to choose to plate both metals simultaneously, if you were to use a potentiostatic method?

(b) (20 points) Assuming 100% of the current goes into plating nickel, from a $NiCl_2$ solution, (100% current efficiency), how many grams of nickel are plated if 3000 amps are applied over 11 minutes? If 400 wrenches were coated with this Ni, at 16 cm^2 area per wrench, how thick is the layer on each wrench (assume that all wrenches are electroplated equally well)?

(c) (15 points) In practice, in a constant current plating bath the current efficiency is NOT 100%. This is because the potential will vary to whatever value is necessary so that electrochemical reactions satisfy the set current density. Describe other chemical reactions that could be involved (in addition to the electroplating reaction). Explain why those other reactions are likely to occur under these conditions.

2. STRIPPING VOLTAMMETRY (50 POINTS)

The amount of metal in aqueous environmental samples can be analyzed using stripping voltammetry. In stripping voltammetry, a metal is deposited (or plated) by applying a reducing potential at an electrode, and then it is stripped back off via oxidation. The number of electrons that it takes to strip off the deposited metal is measured, which is proportional to the concentration of the metal in the sample.

(a) (15 points) A standard additions curve is used to determine the amount of Pb^{2+} in an aqueous environmental sample that has been analyzed with stripping voltammetry. Only two solutions were analyzed: the original sample without any standard added, and the sample containing 5 ppb of standard Pb^{2+} that had been added to it. The number of coulombs that were obtained using stripping voltammetry on these two solutions are 100 μC and 500 μC, respectively. What is the concentration of the Pb^{2+} in the original sample?

(b) (20 points) A different sample, which is saturated with silver carbonate solid is analyzed for the free silver using stripping voltammetry. The amount of Ag^+ determined is 1.27×10^{-4} M. What is the concentration of the Ag^+ in this solution in terms of ppm? What is the K_{sp} of the silver carbonate?

(c) (15 points) Name a different instrumental method (do not include titrations), other than stripping voltammetry, that can be used to analyze very low concentrations of metal in aqueous solutions. How does the method work? How is the metal made detectable (is there any chemistry involved before detection is possible)? What kinds of metals would this method be best suited for detecting?

<u>Useful constants:</u>
F = 96485 Coulombs/mole of electrons
The density of Ni is 8908 kg/m^3
The atomic mass of Ni is 58.6934
The atomic mass of Ag is 107.8682